SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 05 **Total Marks:100**

Annexure A

Exploratory Data Analysis - Based Practical Assessment

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Exploratory Data Analysis-Based Practical Assessment

1. Education Analytics Dataset

Question 1

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Student_ID	Attendance	Study_Hours	Internal_Marks	Result
S01	92	5	45	Pass
S02	55	2	24	Fail
S03	78	4	38	Pass
S04	60	3	28	Fail
S05	88	5	42	Pass
S06	47	1	20	Fail
S07	95	6	48	Pass
S08	72	3	35	Pass
S09	50	2	22	Fail
S10	83	4	40	Pass
S11	66	3	30	Fail
S12	90	5	44	Pass
S13	58	2	25	Fail
S14	76	4	36	Pass
S15	62	3	29	Fail
S16	85	5	41	Pass
S17	45	1	18	Fail
S18	80	4	39	Pass
S19	68	3	32	Pass
S20	53	2	23	Fail

2. Healthcare Analytics Dataset

Question 2

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	125	26	No
P07	52	165	140	30	Yes
P08	35	105	120	24	No
P09	48	155	135	29	Yes
P10	30	95	115	23	No
P11	62	210	155	35	Yes
P12	38	115	122	25	No
P13	50	170	142	30	Yes
P14	27	88	108	21	No

P15	58	190	148	33	Yes
P16	42	130	128	27	No
P17	46	150	132	28	Yes
P18	34	100	116	24	No
P19	53	175	144	31	Yes
P20	29	92	112	22	No

3. Retail Sales Dataset
Question 3
A supermarket wants to analyze product sales based on category, price, quantity sold, discount, and revenue. Perform EDA to identify high-selling products and revenue patterns.

Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
PR01	Grocery	50	20	5	1000
PR02	Snacks	30	35	3	1050
PR03	Dairy	45	18	2	810
PR04	Fruits	60	25	5	1500
PR05	Vegetables	40	30	4	1200
PR06	Grocery	80	15	6	1200
PR07	Snacks	25	40	3	1000
PR08	Dairy	55	16	2	880
PR09	Fruits	70	22	5	1540
PR10	Vegetables	35	28	4	980
PR11	Grocery	90	12	6	1080
PR12	Snacks	20	45	2	900
PR13	Dairy	65	14	3	910
PR14	Fruits	75	20	5	1500
PR15	Vegetables	45	26	4	1170
PR16	Grocery	100	10	7	1000
PR17	Snacks	35	32	3	1120
PR18	Dairy	50	19	2	950
PR19	Fruits	80	18	6	1440
PR20	Vegetables	30	35	3	1050

4. Banking Loan Approval Dataset
Question 4
A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
C01	45000	720	200000	Salaried	Approved
C02	25000	580	150000	Self-employed	Rejected
C03	60000	750	300000	Salaried	Approved
C04	22000	560	120000	Unemployed	Rejected
C05	50000	710	250000	Salaried	Approved
C06	30000	600	180000	Self-employed	Rejected
C07	70000	780	350000	Salaried	Approved
C08	28000	590	160000	Self-employed	Rejected
C09	55000	730	270000	Salaried	Approved
C10	24000	570	130000	Unemployed	Rejected
C11	65000	760	320000	Salaried	Approved
C12	35000	620	200000	Self-employed	Rejected
C13	48000	700	240000	Salaried	Approved

C14	26000	585	140000	Unemployed	Rejected
C15	75000	790	400000	Salaried	Approved
C16	32000	610	190000	Self-employed	Rejected
C17	52000	725	260000	Salaried	Approved
C18	27000	575	150000	Self-employed	Rejected
C19	68000	770	330000	Salaried	Approved
C20	23000	555	125000	Unemployed	Rejected

5. Agriculture Crop Yield Dataset
Question 5
An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
F01	850	28	60	Loamy	3200
F02	600	32	45	Sandy	2100
F03	900	27	65	Loamy	3500
F04	500	34	40	Sandy	1800
F05	780	29	55	Clay	3000
F06	650	31	48	Sandy	2300
F07	920	26	70	Loamy	3700
F08	720	30	52	Clay	2800
F09	550	33	42	Sandy	1900
F10	800	28	58	Loamy	3100
F11	880	27	62	Loamy	3400
F12	620	32	46	Sandy	2200
F13	760	29	54	Clay	2950
F14	940	26	72	Loamy	3800
F15	580	33	44	Sandy	2000
F16	820	28	60	Clay	3150
F17	700	30	50	Clay	2700
F18	960	25	75	Loamy	3900
F19	540	34	41	Sandy	1850
F20	790	29	56	Clay	3050

Common EDA Tasks:

1. Identify the number of rows and columns.
2. Check data types of each column.
3. Find missing values and duplicate values.
4. Calculate mean, median, minimum, maximum, and standard deviation.
5. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.
6. Identify important patterns from the dataset.
7. Write 4 to 5 observations based on EDA.
8. Give a short conclusion from the analysis.

RUBRICS FOR CALCULATION

Exploratory Data Analysis-Based Practical Assessment

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Dataset Understanding	20	Clearly explains dataset source, features, target variables, data types, and problem context with strong clarity	Explains dataset, features, and variables with adequate clarity	Basic explanation of dataset with limited understanding of variables	Poor understanding of dataset; variables and problem context are unclear
Data Cleaning	25	Effectively identifies and handles missing values, duplicates, incorrect entries, and outliers using suitable methods	Handles most missing values, duplicates, and errors correctly	Performs basic cleaning but misses some important data quality issues	Minimal or incorrect data cleaning; major errors remain in the dataset
Data Analysis and Statistical Summary	20	Provides detailed descriptive statistics, comparisons, patterns, and meaningful interpretations	Provides relevant summary statistics and basic interpretation	Provides limited statistics with weak explanation	Lacks statistical analysis or gives incorrect interpretation
Data Visualization	20	Uses highly appropriate charts with proper labels, titles, and clear visual interpretation	Uses suitable charts with mostly clear labels and explanation	Uses limited or basic charts with unclear interpretation	Poor or incorrect visualizations; charts do not support analysis
Insight Generation and Reporting	15	Presents strong insights, conclusions, and recommendations based on data evidence	Presents clear findings and conclusions with reasonable support	Provides basic findings but lacks depth or proper justification	Findings are unclear, unsupported, or not related to the data

CO1-ASSESSMENT TOOL -2 Concept Map

QUESTIONS : 5 **MARKS : 50**

Q No	Dataset Understanding (20%)	Data Cleaning (25%)	Data Analysis and Statistical Summary (20%)	Data Visualization (20%)	Insight Generation and Reporting (15%)	Total Score (100%)
1	/2.5	/2.5	/2.0	/2.0	/1.0	/10
2	/2.5	/2.5	/2.0	/2.0	/1.0	/10
3	/2.5	/2.5	/2.0	/2.0	/1.0	/10
4	/2.5	/2.5	/2.0	/2.0	/1.0	/10
5	/2.5	/2.5	/2.0	/2.0	/1.0	/10
Overall Rubrics Value	/12.5	/12.5	/10	/10	/10	/50
	/25	/25	/20	/20	/10	/100

In [18]:

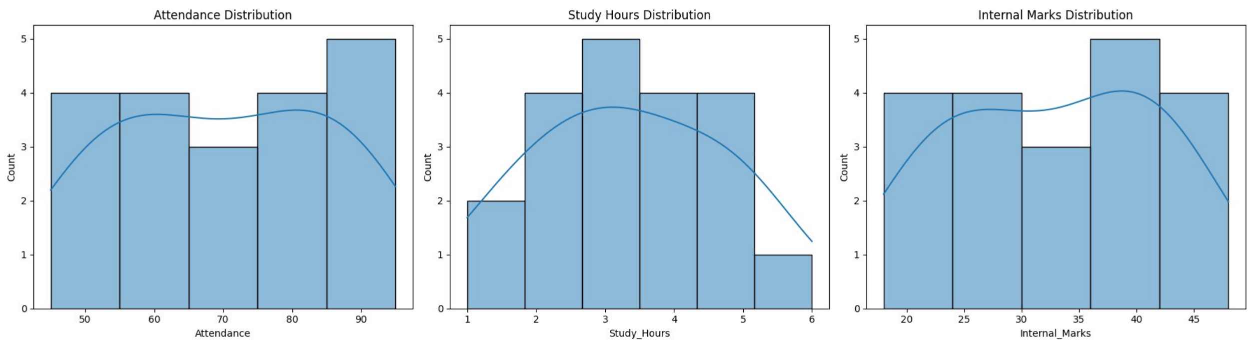
```
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
data = {
        "Student_ID":["S01","S02","S03","S04","S05","S06","S07","S08","S09","S10",
                      "S11","S12","S13","S14","S15","S16","S17","S18","S19","S20"]

    "Attendance":[92,55,78,60,88,47,95,72,50,83,
                  66,90,58,76,62,85,45,80,68,53],

    "Study_Hours":[5,2,4,3,5,1,6,3,2,4,
                   3,5,2,4,3,5,1,4,3,2],

    "Internal_Marks":[45,24,38,28,42,20,48,35,22,40,
                      30,44,25,36,29,41,18,39,32,23],

        "Result":["Pass","Fail","Pass","Fail","Pass","Fail","Pass","Pass","Fail","
                  "Fail","Pass","Fail","Pass","Fail","Pass","Fail","Pass","Pass","
}

df = pd.DataFrame(data)

# Display Dataset
print("Education Analytics Dataset")
print(df)

# 1. Number of Rows and Columns
print("\nRows and Columns:", df.shape)

# 2. Data Types
print("\nData Types")
print(df.dtypes)

# 3. Missing Values
print("\nMissing Values")
print(df.isnull().sum())

# Duplicate Values
print("\nDuplicate Values:", df.duplicated().sum())

# 4. Statistical Measures
print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))
```

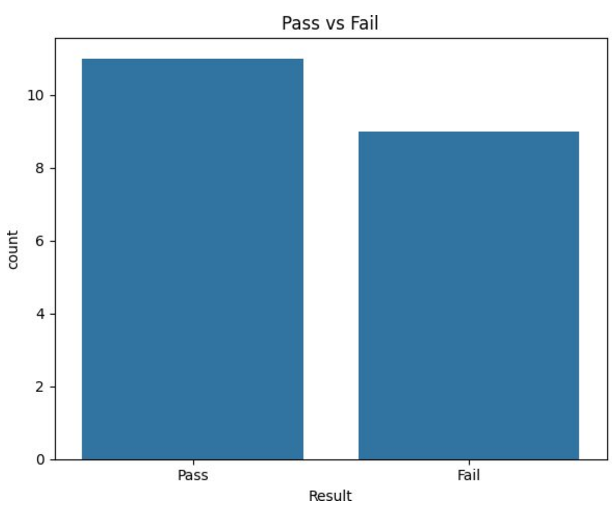
```python
print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

# -------------------- Charts --------------------

# Histograms for Attendance, Study_Hours, Internal_Marks
fig, axes = plt.subplots(1, 3, figsize=(18, 5))
sns.histplot(df["Attendance"], bins=5, kde=True, ax=axes[0])
axes[0].set_title("Attendance Distribution")
sns.histplot(df["Study_Hours"], bins=6, kde=True, ax=axes[1])
axes[1].set_title("Study Hours Distribution")
sns.histplot(df["Internal_Marks"], bins=5, kde=True, ax=axes[2])
axes[2].set_title("Internal Marks Distribution")
plt.tight_layout()
plt.show()

# Countplot for Result and Boxplot for Attendance vs Result
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.countplot(x="Result", data=df, ax=axes[0])
axes[0].set_title("Pass vs Fail")
sns.boxplot(x="Result", y="Attendance", data=df, ax=axes[1])
axes[1].set_title("Attendance vs Result")
plt.tight_layout()
plt.show()

# Boxplots for Study_Hours vs Result and Internal_Marks vs Result
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.boxplot(x="Result", y="Study_Hours", data=df, ax=axes[0])
axes[0].set_title("Study  Hours  vs  Result")
sns.boxplot(x="Result", y="Internal_Marks", data=df, ax=axes[1])
axes[1].set_title("Internal Marks vs Result")
plt.tight_layout()
plt.show()

# Scatter Plot for Study_Hours vs Internal_Marks and Correlation Heatmap
fig, axes = plt.subplots(1, 2, figsize=(14, 6)) # Increased figsize for better
sns.scatterplot(x="Study_Hours", y="Internal_Marks", hue="Result", data=df, ax
axes[0].set_title("Study Hours vs Internal Marks")

numeric_df = df[["Attendance","Study_Hours","Internal_Marks"]]
sns.heatmap(numeric_df.corr(), annot=True, cmap="Blues", ax=axes[1])
axes[1].set_title("Correlation Matrix")
plt.tight_layout()
plt.show()

# 5. Important Patterns
print("\nImportant Patterns")
print("• Students with higher attendance mostly passed.")
print("• Students who studied more hours scored higher marks.")
print("• Higher internal marks are associated with passing.")
```

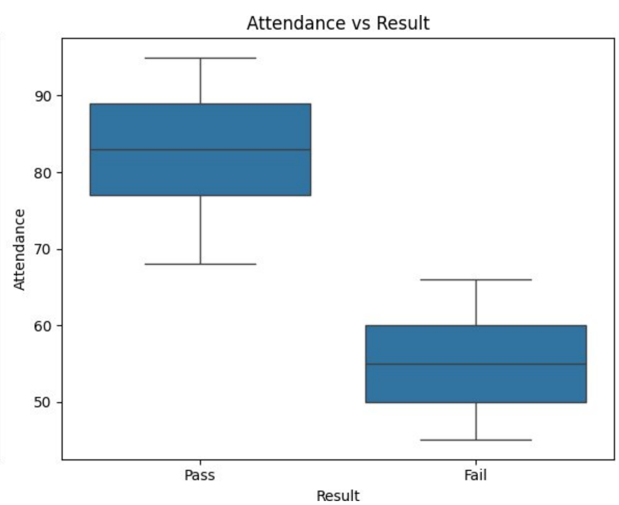
```python
print("• Attendance and Internal Marks have a positive correlation.")

# 6. Observations
print("\nObservations")
print("1. Higher attendance improves the chances of passing.")
print("2. Students studying 4-6 hours scored better.")
print("3. Internal marks are higher for students who passed.")
print("4. Attendance and study hours positively affect performance.")
print("5. Internal Marks are the strongest indicator of the final result.")

# 7. Conclusion
print("\nConclusion")
print("The analysis shows that Attendance, Study Hours, and Internal Marks sig
```

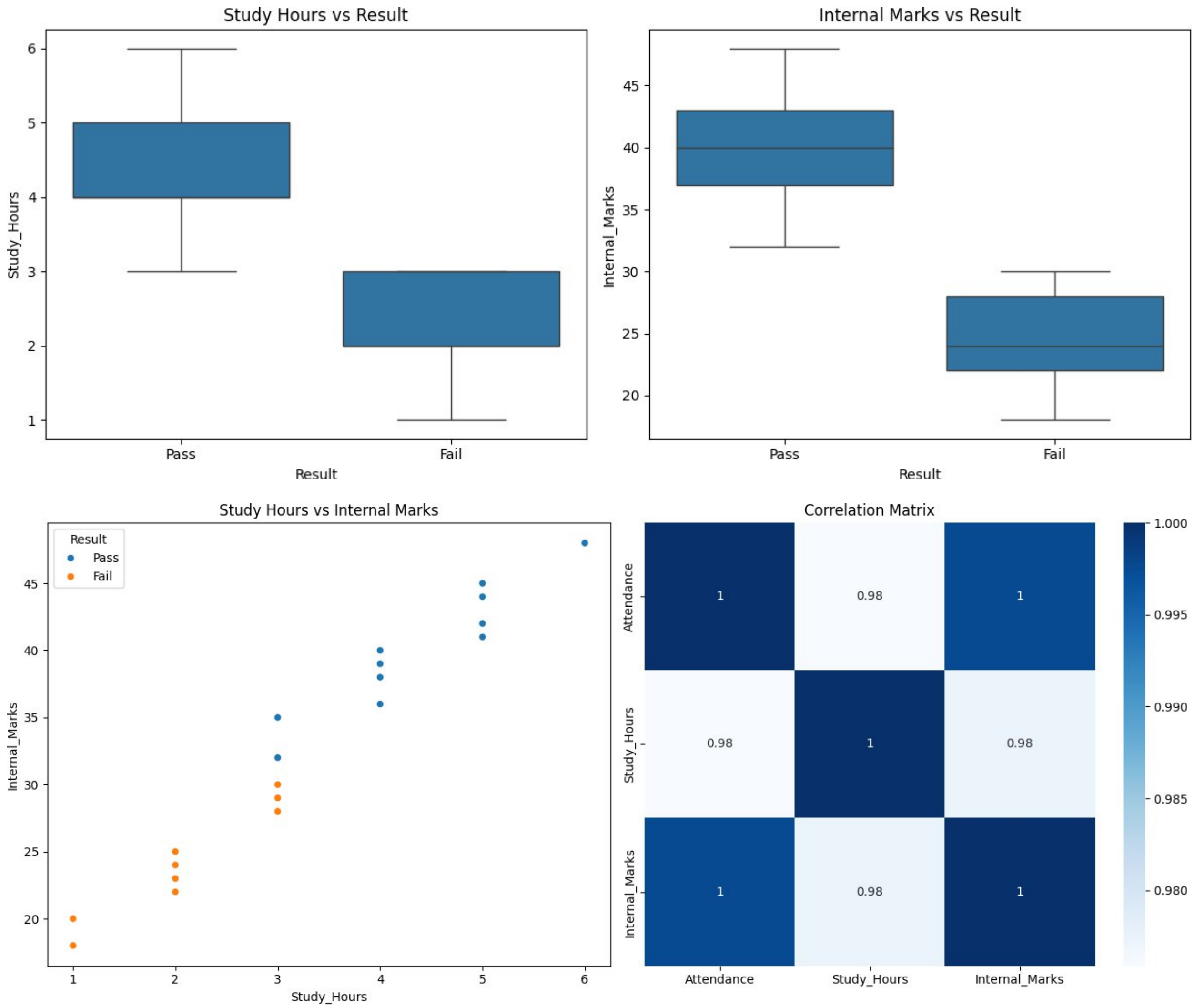
```
Education Analytics Dataset
   Student_ID  Attendance  Study_Hours  Internal_Marks Result
0         S01          92            5              45   Pass
1         S02          55            2              24   Fail
2         S03          78            4              38   Pass
3         S04          60            3              28   Fail
4         S05          88            5              42   Pass
5         S06          47            1              20   Fail
6         S07          95            6              48   Pass
7         S08          72            3              35   Pass
8         S09          50            2              22   Fail
9         S10          83            4              40   Pass
10        S11          66            3              30   Fail
11        S12          90            5              44   Pass
12        S13          58            2              25   Fail
13        S14          76            4              36   Pass
14        S15          62            3              29   Fail
15        S16          85            5              41   Pass
16        S17          45            1              18   Fail
17        S18          80            4              39   Pass
18        S19          68            3              32   Pass
19        S20          53            2              23   Fail

Rows and Columns: (20, 5)

Data Types
Student_ID        object
Attendance         int64
Study_Hours        int64
Internal_Marks     int64
Result            object
dtype: object

Missing Values
Student_ID        0
Attendance        0
Study_Hours       0
Internal_Marks    0
Result            0
dtype: int64

Duplicate Values:  0

Mean
Attendance        70.15
Study_Hours        3.35
Internal_Marks    32.95
dtype: float64

Median
Attendance        70.0
Study_Hours        3.0
Internal_Marks    33.5
dtype: float64
```

```
Minimum
Attendance        45
Study_Hours        1
Internal_Marks    18
dtype: int64

Maximum
Attendance        95
Study_Hours        6
Internal_Marks    48
dtype: int64

Standard Deviation
Attendance        15.938286
Study_Hours        1.424411
Internal_Marks     9.087151
dtype: float64
```









Important Patterns
• Students with higher attendance mostly passed.
• Students who studied more hours scored higher marks.
• Higher internal marks are associated with passing.
• Attendance and Internal Marks have a positive correlation.

Observations
1. Higher attendance improves the chances of passing.
2. Students studying 4–6 hours scored better.
3. Internal marks are higher for students who passed.
4. Attendance and study hours positively affect performance.
5. Internal Marks are the strongest indicator of the final result.

Conclusion
The analysis shows that Attendance, Study Hours, and Internal Marks significant ly influence student performance. Students with higher attendance and study hou rs generally achieve better results.

The correlation matrix and pair plot are presented individually as they provide distinct analytical views. A pair plot inherently generates a grid of relationships, making it a comprehensive multi-plot figure on its own.

In [13]:

```python
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
data = {
        "Patient_ID":["P01","P02","P03","P04","P05","P06","P07","P08","P09","P10",
                      "P11","P12","P13","P14","P15","P16","P17","P18","P19","P20"]

    "Age":[45,32,55,28,60,40,52,35,48,30,
           62,38,50,27,58,42,46,34,53,29],

    "Sugar_Level":[145,98,180,90,200,120,165,105,155,95,
                   210,115,170,88,190,130,150,100,175,92],

    "BP":[130,118,145,110,150,125,140,120,135,115,
          155,122,142,108,148,128,132,116,144,112],

    "BMI":[28,23,31,22,34,26,30,24,29,23,
           35,25,30,21,33,27,28,24,31,22],

        "Disease_Status":["Yes","No","Yes","No","Yes","No","Yes","No","Yes","No",
                          "Yes","No","Yes","No","Yes","No","Yes","No","Yes","No"]
}

df = pd.DataFrame(data)

# Display Dataset
print("Healthcare Analytics Dataset")
print(df)

# 1. Number of Rows and Columns
print("\nRows and Columns:", df.shape)

# 2. Data Types
print("\nData Types")
print(df.dtypes)

# 3. Missing Values
print("\nMissing Values")
print(df.isnull().sum())

# Duplicate Values
print("\nDuplicate Values:", df.duplicated().sum())

# 4. Statistical Measures
print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))
```

```
print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

# 5. Charts

# Histogram and Bar Chart
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.histplot(df["Sugar_Level"], bins=5, kde=True, color="skyblue", ax=axes[0])
axes[0].set_title("Sugar Level Distribution")
axes[0].set_xlabel("Sugar Level")
axes[0].set_ylabel("Frequency")
sns.countplot(x="Disease_Status", data=df, ax=axes[1])
axes[1].set_title("Disease Status Count")
plt.tight_layout()
plt.show()

# Scatter Plot and Box Plot
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.scatterplot(x="Sugar_Level", y="BMI", hue="Disease_Status", data=df, ax=ax
axes[0].set_title("Sugar Level vs BMI")
sns.boxplot(x="Disease_Status", y="Sugar_Level", data=df, ax=axes[1])
axes[1].set_title("Sugar Level by Disease Status")
plt.tight_layout()
plt.show()

# 6. Important Patterns
print("\nImportant Patterns")
print("• Patients with higher sugar levels are mostly affected by disease.")
print("• Higher BMI is commonly observed in diseased patients.")
print("• Patients with high BP also tend to have disease.")
print("• Lower sugar levels are mostly associated with healthy patients.")

# 7. Observations
print("\nObservations")
print("1. High sugar level increases disease risk.")
print("2. Patients with higher BMI are more likely to have disease.")
print("3. High BP is common among diseased patients.")
print("4. Age has a moderate effect on disease.")
print("5. Sugar Level is the strongest indicator of disease.")

# 8. Conclusion
print("\nConclusion")
print("The EDA shows that Sugar Level, BMI, and BP are the major factors assoc
```

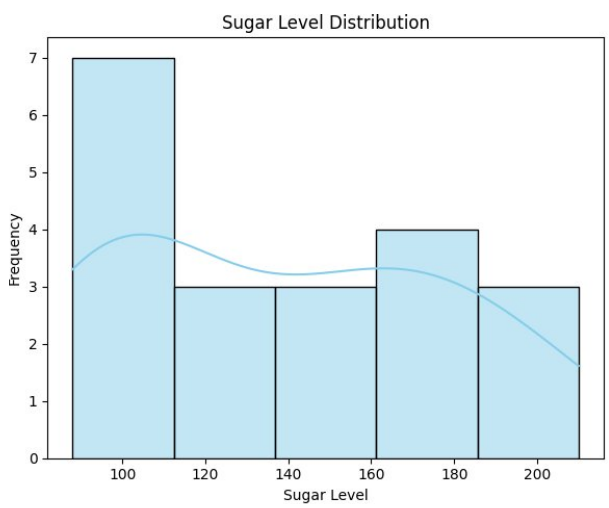
Healthcare Analytics Dataset

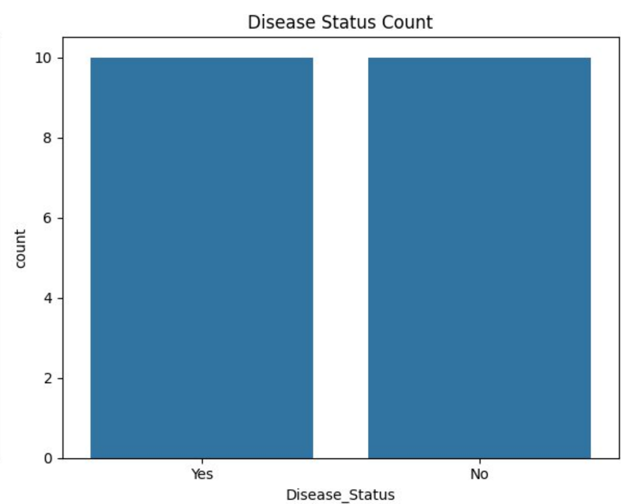
	Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
0	P01	45	145	130	28	Yes
1	P02	32	98	118	23	No
2	P03	55	180	145	31	Yes
3	P04	28	90	110	22	No
4	P05	60	200	150	34	Yes
5	P06	40	120	125	26	No
6	P07	52	165	140	30	Yes
7	P08	35	105	120	24	No
8	P09	48	155	135	29	Yes
9	P10	30	95	115	23	No
10	P11	62	210	155	35	Yes
11	P12	38	115	122	25	No
12	P13	50	170	142	30	Yes
13	P14	27	88	108	21	No
14	P15	58	190	148	33	Yes
15	P16	42	130	128	27	No
16	P17	46	150	132	28	Yes
17	P18	34	100	116	24	No
18	P19	53	175	144	31	Yes
19	P20	29	92	112	22	No

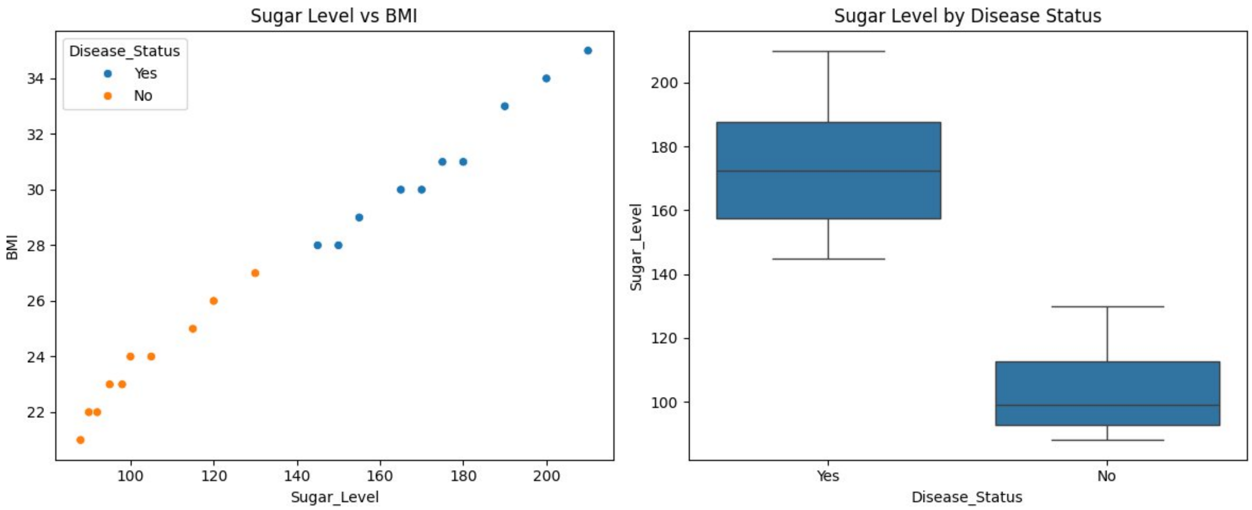
Rows and Columns: (20, 6)

```
Data Types
Patient_ID         object
Age                 int64
Sugar_Level         int64
BP                  int64
BMI                 int64
Disease_Status     object
dtype: object

Missing Values
Patient_ID        0
Age               0
Sugar_Level       0
BP                0
BMI               0
Disease_Status    0
dtype: int64

Duplicate Values: 0

Mean
Age              43.20
Sugar_Level     138.65
BP              129.75
BMI              27.30
dtype: float64

Median
Age              43.5
```

```
Sugar_Level     137.5
BP              129.0
BMI              27.5
dtype: float64

Minimum
Age              27
Sugar_Level      88
BP              108
BMI              21
dtype: int64

Maximum
Age              62
Sugar_Level     210
BP              155
BMI              35
dtype: int64

Standard Deviation
Age             11.316499
Sugar_Level     40.334490
BP              14.523574
BMI              4.256265
dtype: float64
```







Important Patterns
• Patients with higher sugar levels are mostly affected by disease.
• Higher BMI is commonly observed in diseased patients.
• Patients with high BP also tend to have disease.
• Lower sugar levels are mostly associated with healthy patients.

Observations
1. High sugar level increases disease risk.
2. Patients with higher BMI are more likely to have disease.
3. High BP is common among diseased patients.
4. Age has a moderate effect on disease.
5. Sugar Level is the strongest indicator of disease.

Conclusion
The EDA shows that Sugar Level, BMI, and BP are the major factors associated wi th disease status. Patients with higher values are more likely to have the dise ase.

In [14]:

```
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
data = {
        "Product_ID":["PR01","PR02","PR03","PR04","PR05","PR06","PR07","PR08","PR0
                      "PR11","PR12","PR13","PR14","PR15","PR16","PR17","PR18","PR1

        "Category":["Grocery","Snacks","Dairy","Fruits","Vegetables",
                    "Grocery","Snacks","Dairy","Fruits","Vegetables",
                    "Grocery","Snacks","Dairy","Fruits","Vegetables",
                  "Grocery","Snacks","Dairy","Fruits","Vegetables"],

    "Price":[50,30,45,60,40,80,25,55,70,35,
             90,20,65,75,45,100,35,50,80,30],

    "Quantity_Sold":[20,35,18,25,30,15,40,16,22,28,
                     12,45,14,20,26,10,32,19,18,35],
```

```
    "Discount":[5,3,2,5,4,6,3,2,5,4,
                6,2,3,5,4,7,3,2,6,3],

    "Revenue":[1000,1050,810,1500,1200,1200,1000,880,1540,980,
               1080,900,910,1500,1170,1000,1120,950,1440,1050]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display Dataset
print("Retail Sales Dataset")
print(df)

# 1. Number of Rows and Columns
print("\nRows and Columns:", df.shape)

# 2. Check Data Types
print("\nData  Types")
print(df.dtypes)

# 3. Missing Values
print("\nMissing Values")
print(df.isnull().sum())

# Duplicate Values
print("\nDuplicate Values:", df.duplicated().sum())

# 4. Statistical Measures
print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

# ----------------- Charts -----------------

# 1. Histogram and Bar Chart
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
sns.histplot(df["Revenue"], bins=5, kde=True, ax=axes[0])
axes[0].set_title("Revenue Distribution")

category_revenue = df.groupby("Category")["Revenue"].sum()
```

```
category_revenue.plot(kind="bar", color="skyblue", ax=axes[1])
axes[1].set_title("Revenue by Category")
axes[1].set_xlabel("Category")
axes[1].set_ylabel("Revenue")
plt.tight_layout()
plt.show()

# 3. Scatter Plot and Box Plot
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
sns.scatterplot(x="Price", y="Revenue", hue="Category", data=df, ax=axes[0])
axes[0].set_title("Price vs Revenue")
sns.boxplot(x="Category", y="Revenue", data=df, ax=axes[1])
axes[1].set_title("Revenue by Category")
plt.tight_layout()
plt.show()

# 5. Important Patterns
print("\nImportant  Patterns")
print("• Fruits category has the highest revenue.")
print("• Snacks category has the highest quantity sold.")
print("• Higher-priced products generally earn more revenue.")
print("• Discount has less impact on revenue than price.")

# 6. Observations
print("\nObservations")
print("1. Fruits category generated the highest revenue.")
print("2. Snacks products recorded the highest sales.")
print("3.  Revenue  increases  with  product  price.")
print("4. Discount has only a small effect on revenue.")
print("5. Price and Quantity Sold are the main factors affecting revenue.")

# 7. Conclusion
print("\nConclusion")
print("The EDA shows that Price and Quantity Sold significantly influence Reve
```

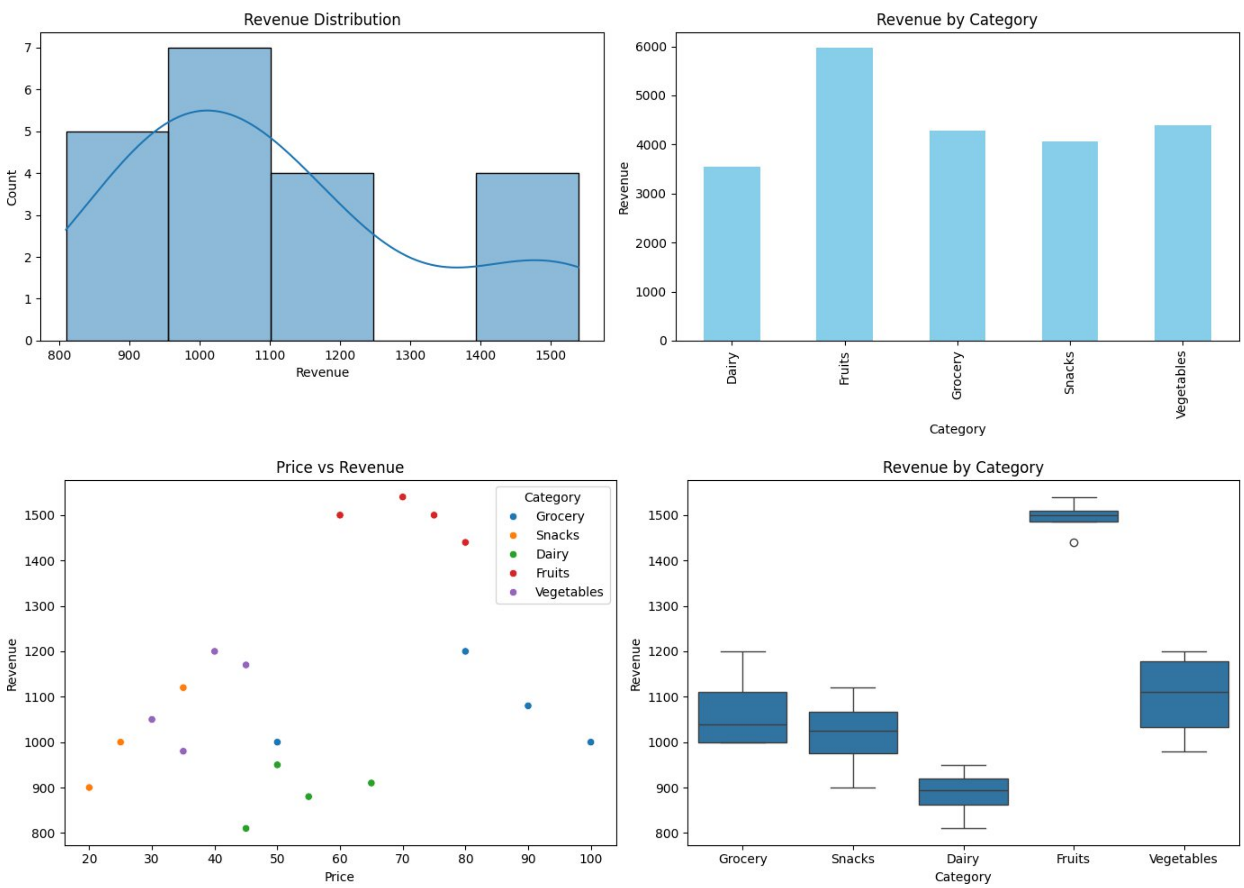
Retail Sales Dataset

	Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
0	PR01	Grocery	50	20	5	1000
1	PR02	Snacks	30	35	3	1050
2	PR03	Dairy	45	18	2	810
3	PR04	Fruits	60	25	5	1500
4	PR05	Vegetables	40	30	4	1200
5	PR06	Grocery	80	15	6	1200
6	PR07	Snacks	25	40	3	1000
7	PR08	Dairy	55	16	2	880
8	PR09	Fruits	70	22	5	1540
9	PR10	Vegetables	35	28	4	980
10	PR11	Grocery	90	12	6	1080
11	PR12	Snacks	20	45	2	900
12	PR13	Dairy	65	14	3	910
13	PR14	Fruits	75	20	5	1500
14	PR15	Vegetables	45	26	4	1170
15	PR16	Grocery	100	10	7	1000
16	PR17	Snacks	35	32	3	1120
17	PR18	Dairy	50	19	2	950
18	PR19	Fruits	80	18	6	1440
19	PR20	Vegetables	30	35	3	1050

Rows and Columns: (20, 6)

```
Data Types
Product_ID        object
Category          object
Price              int64
Quantity_Sold      int64
Discount           int64
Revenue            int64
dtype: object

Missing Values
Product_ID        0
Category          0
Price             0
Quantity_Sold     0
Discount          0
Revenue           0
dtype: int64

Duplicate Values:  0

Mean
Price               54.0
Quantity_Sold       24.0
Discount             4.0
Revenue           1114.0
dtype: float64

Median
Price               50.0
```

```
Quantity_Sold       21.0
Discount             4.0
Revenue           1050.0
dtype: float64

Minimum
Price             20
Quantity_Sold     10
Discount           2
Revenue          810
dtype: int64

Maximum
Price             100
Quantity_Sold      45
Discount            7
Revenue          1540
dtype: int64

Standard Deviation
Price             22.803509
Quantity_Sold      9.673621
Discount           1.555973
Revenue          221.416493
dtype: float64
```



Important Patterns
• Fruits category has the highest revenue.
• Snacks category has the highest quantity sold.
• Higher-priced products generally earn more revenue.
• Discount has less impact on revenue than price.

Observations
1. Fruits category generated the highest revenue.
2. Snacks products recorded the highest sales.
3. Revenue increases with product price.
4. Discount has only a small effect on revenue.
5. Price and Quantity Sold are the main factors affecting revenue.

Conclusion
The EDA shows that Price and Quantity Sold significantly influence Revenue. Fruits are the most profitable category, while Snacks have the highest sales volume.

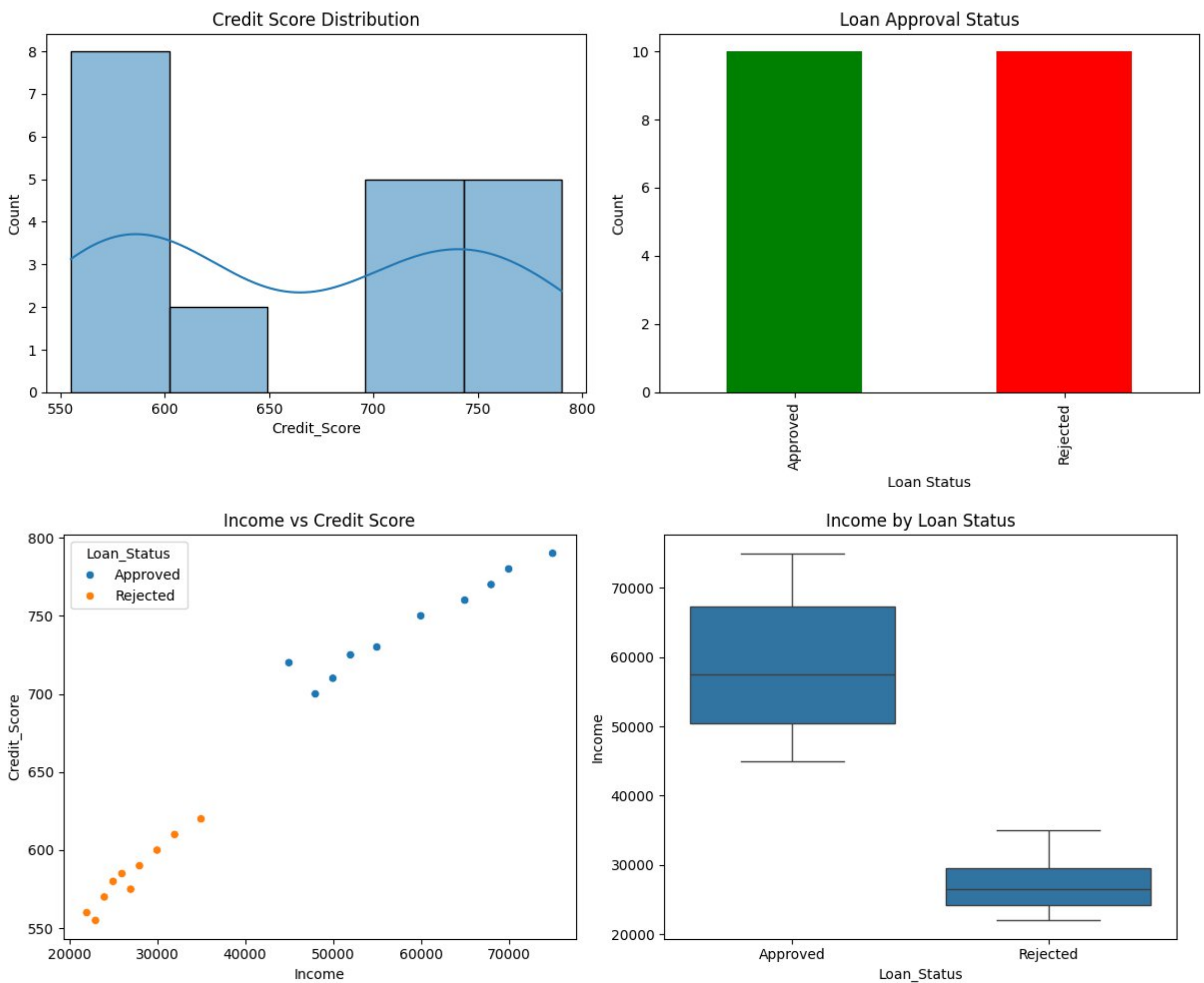
In [15]:
```
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
data = {
        "Customer_ID":["C01","C02","C03","C04","C05","C06","C07","C08","C09","C10"
                       "C11","C12","C13","C14","C15","C16","C17","C18","C19","C20"

    "Income":[45000,25000,60000,22000,50000,30000,70000,28000,55000,24000,
              65000,35000,48000,26000,75000,32000,52000,27000,68000,23000],

    "Credit_Score":[720,580,750,560,710,600,780,590,730,570,
                    760,620,700,585,790,610,725,575,770,555],

    "Loan_Amount":[200000,150000,300000,120000,250000,180000,350000,160000,270
                   320000,200000,240000,140000,400000,190000,260000,150000,330

    "Employment_Type":["Salaried","Self-employed","Salaried","Unemployed","Sal
                       "Self-employed","Salaried","Self-employed","Salaried","
                       "Salaried","Self-employed","Salaried","Unemployed","Sal
                       "Self-employed","Salaried","Self-employed","Salaried","

    "Loan_Status":["Approved","Rejected","Approved","Rejected","Approved",
                   "Rejected","Approved","Rejected","Approved","Rejected",
                   "Approved","Rejected","Approved","Rejected","Approved",
                   "Rejected","Approved","Rejected","Approved","Rejected"]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display Dataset
print("Banking Loan Approval Dataset")
print(df)
```

```python
# 1. Number of Rows and Columns
print("\nRows and Columns:", df.shape)

# 2. Data Types
print("\nData Types")
print(df.dtypes)

# 3. Missing Values
print("\nMissing Values")
print(df.isnull().sum())

# Duplicate Values
print("\nDuplicate Values:", df.duplicated().sum())

# 4. Statistical Measures
print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

# -------------------- Charts --------------------

# 1. Histogram and Bar Chart
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.histplot(df["Credit_Score"], bins=5, kde=True, ax=axes[0])
axes[0].set_title("Credit Score Distribution")

loan_count = df["Loan_Status"].value_counts()
loan_count.plot(kind="bar", color=["green","red"], ax=axes[1])
axes[1].set_title("Loan Approval Status")
axes[1].set_xlabel("Loan Status")
axes[1].set_ylabel("Count")
plt.tight_layout()
plt.show()

# 3. Scatter Plot and Box Plot
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.scatterplot(x="Income", y="Credit_Score", hue="Loan_Status", data=df, ax=a
axes[0].set_title("Income vs Credit Score")
sns.boxplot(x="Loan_Status", y="Income", data=df, ax=axes[1])
axes[1].set_title("Income by Loan Status")
plt.tight_layout()
```

```
plt.show()

# 5. Important Patterns
print("\nImportant Patterns")
print("• Customers with higher income are mostly approved.")
print("• Higher credit scores increase loan approval chances.")
print("• Approved customers generally have higher income.")
print("• Loan approval mainly depends on income and credit score.")

# 6. Observations
print("\nObservations")
print("1. Customers with high income are more likely to get loans.")
print("2. Higher credit scores lead to higher approval rates.")
print("3. Rejected customers usually have lower income.")
print("4. Income and Credit Score have a positive relationship.")
print("5. Credit Score is an important factor in loan approval.")

# 7. Conclusion
print("\nConclusion")
print("The analysis shows that Income and Credit Score play a major role in lo
```

```
Banking Loan Approval Dataset
   Customer_ID  Income  Credit_Score  Loan_Amount Employment_Type Loan_Status
0          C01   45000           720       200000        Salaried    Approved
1          C02   25000           580       150000   Self-employed    Rejected
2          C03   60000           750       300000        Salaried    Approved
3          C04   22000           560       120000      Unemployed    Rejected
4          C05   50000           710       250000        Salaried    Approved
5          C06   30000           600       180000   Self-employed    Rejected
6          C07   70000           780       350000        Salaried    Approved
7          C08   28000           590       160000   Self-employed    Rejected
8          C09   55000           730       270000        Salaried    Approved
9          C10   24000           570       130000      Unemployed    Rejected
10         C11   65000           760       320000        Salaried    Approved
11         C12   35000           620       200000   Self-employed    Rejected
12         C13   48000           700       240000        Salaried    Approved
13         C14   26000           585       140000      Unemployed    Rejected
14         C15   75000           790       400000        Salaried    Approved
15         C16   32000           610       190000   Self-employed    Rejected
16         C17   52000           725       260000        Salaried    Approved
17         C18   27000           575       150000   Self-employed    Rejected
18         C19   68000           770       330000        Salaried    Approved
19         C20   23000           555       125000      Unemployed    Rejected

Rows and Columns: (20, 6)

Data Types
Customer_ID        object
Income              int64
Credit_Score        int64
Loan_Amount         int64
Employment_Type    object
Loan_Status  object dtype:
object

Missing Values
Customer_ID        0
Income             0
Credit_Score       0
Loan_Amount        0
Employment_Type    0
Loan_Status        0
dtype: int64

Duplicate Values: 0

Mean
Income            43000.0
Credit_Score        664.0
Loan_Amount      223250.0
dtype: float64

Median
Income            40000.0
Credit_Score        660.0
```

```
Loan_Amount      200000.0
dtype: float64

Minimum
Income            22000
Credit_Score        555
Loan_Amount      120000
dtype: int64

Maximum
Income            75000
Credit_Score        790
Loan_Amount      400000
dtype: int64

Standard Deviation
Income           17923.815383
Credit_Score        85.526235
Loan_Amount      83733.207272
dtype: float64
```



Important Patterns
• Customers with higher income are mostly approved.
• Higher credit scores increase loan approval chances.
• Approved customers generally have higher income.
• Loan approval mainly depends on income and credit score.

Observations
1. Customers with high income are more likely to get loans.
2. Higher credit scores lead to higher approval rates.
3. Rejected customers usually have lower income.
4. Income and Credit Score have a positive relationship.
5. Credit Score is an important factor in loan approval.

Conclusion
The analysis shows that Income and Credit Score play a major role in loan approval. Customers with higher values have a greater chance of getting their loans approved.

In [16]:

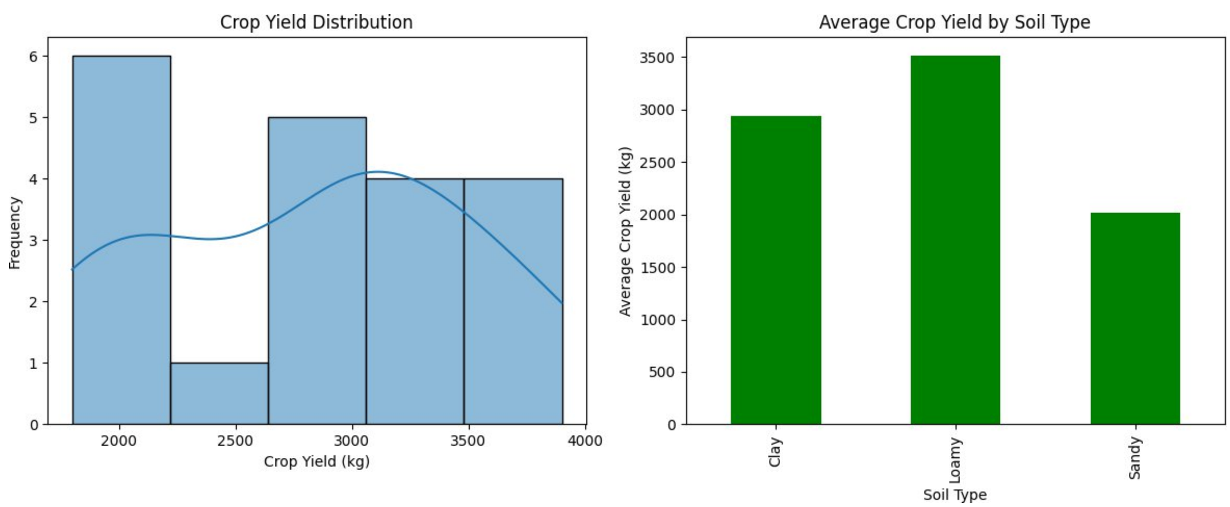
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
    "Farm_ID":["F01","F02","F03","F04","F05","F06","F07","F08","F09","F10",
               "F11","F12","F13","F14","F15","F16","F17","F18","F19","F20"],

    "Rainfall_mm":[850,600,900,500,780,650,920,720,550,800,
                   880,620,760,940,580,820,700,960,540,790],

    "Temperature":[28,32,27,34,29,31,26,30,33,28,
                   27,32,29,26,33,28,30,25,34,29],

    "Fertilizer_kg":[60,45,65,40,55,48,70,52,42,58,
                     62,46,54,72,44,60,50,75,41,56],

      "Soil_Type":["Loamy","Sandy","Loamy","Sandy","Clay",
                   "Sandy","Loamy","Clay","Sandy","Loamy",
                   "Loamy","Sandy","Clay","Loamy","Sandy",
                 "Clay","Clay","Loamy","Sandy","Clay"],

    "Crop_Yield_kg":[3200,2100,3500,1800,3000,2300,3700,2800,1900,3100,
                     3400,2200,2950,3800,2000,3150,2700,3900,1850,3050]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display Dataset
print("Agriculture Crop Yield Dataset")
print(df)

# 1. Number of Rows and Columns
print("\nRows and Columns:", df.shape)

# 2. Data Types
```

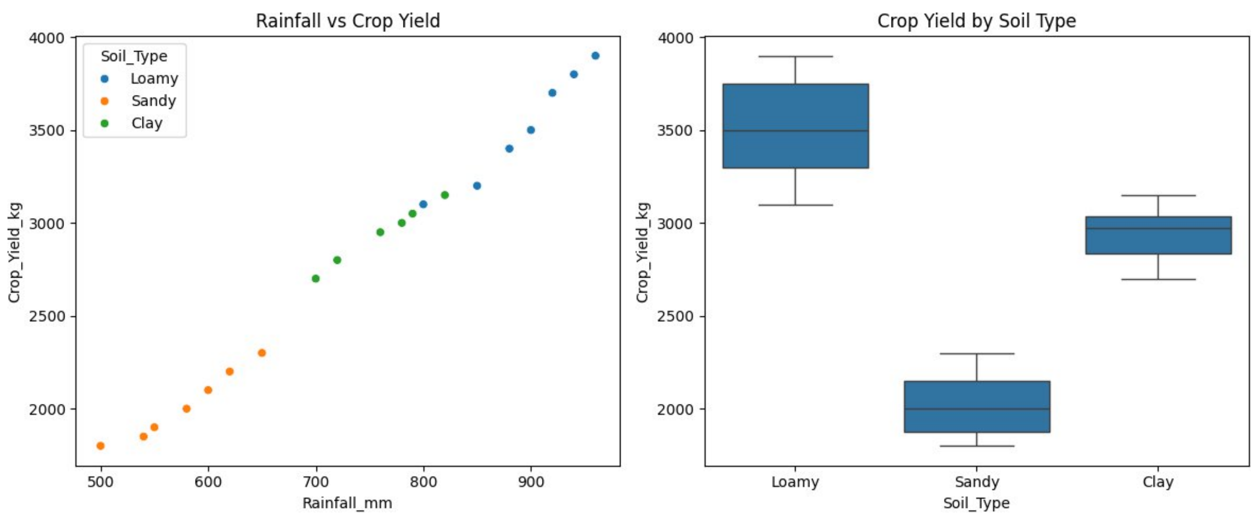
```python
print("\nData Types")
print(df.dtypes)

# 3. Missing Values
print("\nMissing Values")
print(df.isnull().sum())

# Duplicate Values
print("\nDuplicate Values:", df.duplicated().sum())

# 4. Statistical Measures
print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

# ------------------ Charts ------------------

# 1. Histogram and Bar Chart
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.histplot(df["Crop_Yield_kg"], bins=5, kde=True, ax=axes[0])
axes[0].set_title("Crop Yield Distribution")
axes[0].set_xlabel("Crop Yield (kg)")
axes[0].set_ylabel("Frequency")

soil_yield = df.groupby("Soil_Type")["Crop_Yield_kg"].mean()
soil_yield.plot(kind="bar", color="green", ax=axes[1])
axes[1].set_title("Average Crop Yield by Soil Type")
axes[1].set_xlabel("Soil Type")
axes[1].set_ylabel("Average Crop Yield (kg)")
plt.tight_layout()
plt.show()

# 3. Scatter Plot and Box Plot
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
sns.scatterplot(x="Rainfall_mm", y="Crop_Yield_kg", hue="Soil_Type", data=df,
axes[0].set_title("Rainfall vs Crop Yield")
sns.boxplot(x="Soil_Type", y="Crop_Yield_kg", data=df, ax=axes[1])
axes[1].set_title("Crop Yield by Soil Type")
plt.tight_layout()
plt.show()

print("\nImportant Patterns")
```

```
print("• Higher rainfall generally results in higher crop yield.")
print("• Loamy soil produces better crop yield than Sandy soil.")
print("• Higher fertilizer usage is associated with higher crop production.")
print("• Temperature has a moderate impact on crop yield.")

print("\nObservations")
print("1. Farms with higher rainfall produce higher crop yield.")
print("2. Loamy soil has the highest average crop yield.")
print("3. Sandy soil gives comparatively lower crop yield.")
print("4. Fertilizer usage positively influences crop production.")
print("5. Rainfall and fertilizer are the major factors affecting crop yield."
print("\nConclusion")
print("The analysis shows that rainfall, soil type, and fertilizer usage signi
```

```
Agriculture Crop Yield Dataset
   Farm_ID  Rainfall_mm  Temperature  Fertilizer_kg Soil_Type  Crop_Yield_kg
0      F01          850           28             60     Loamy           3200
1      F02          600           32             45     Sandy           2100
2      F03          900           27             65     Loamy           3500
3      F04          500           34             40     Sandy           1800
4      F05          780           29             55      Clay           3000
5      F06          650           31             48     Sandy           2300
6      F07          920           26             70     Loamy           3700
7      F08          720           30             52      Clay           2800
8      F09          550           33             42     Sandy           1900
9      F10          800           28             58     Loamy           3100
10     F11          880           27             62     Loamy           3400
11     F12          620           32             46     Sandy           2200
12     F13          760           29             54      Clay           2950
13     F14          940           26             72     Loamy           3800
14     F15          580           33             44     Sandy           2000
15     F16          820           28             60      Clay           3150
16     F17          700           30             50      Clay           2700
17     F18          960           25             75     Loamy           3900
18     F19          540           34             41     Sandy           1850
19     F20          790           29             56      Clay           3050

Rows and Columns:  (20, 6)

Data Types
Farm_ID           object
Rainfall_mm        int64
Temperature        int64
Fertilizer_kg      int64
Soil_Type         object
Crop_Yield_kg      int64
dtype: object

Missing Values
Farm_ID          0
Rainfall_mm      0
Temperature      0
Fertilizer_kg    0
Soil_Type        0
Crop_Yield_kg    0
dtype: int64

Duplicate Values:  0

Mean
Rainfall_mm        743.00
Temperature         29.55
Fertilizer_kg       54.75
Crop_Yield_kg     2820.00
dtype: float64

Median
Rainfall_mm        770.0
```

```
Temperature          29.0
Fertilizer_kg        54.5
Crop_Yield_kg      2975.0
dtype: float64

Minimum
Rainfall_mm         500
Temperature          25
Fertilizer_kg        40
Crop_Yield_kg      1800
dtype: int64

Maximum
Rainfall_mm         960
Temperature          34
Fertilizer_kg        75
Crop_Yield_kg      3900
dtype: int64

Standard Deviation
Rainfall_mm        144.335136
Temperature          2.762055
Fertilizer_kg       10.507516
Crop_Yield_kg      681.407213
dtype: float64
```





Important Patterns

- Higher rainfall generally results in higher crop yield.
- Loamy soil produces better crop yield than Sandy soil.
- Higher fertilizer usage is associated with higher crop production.
- Temperature has a moderate impact on crop yield.

Observations

1. Farms with higher rainfall produce higher crop yield.
2. Loamy soil has the highest average crop yield.
3. Sandy soil gives comparatively lower crop yield.
4. Fertilizer usage positively influences crop production.
5. Rainfall and fertilizer are the major factors affecting crop yield.

Conclusion

The analysis shows that rainfall, soil type, and fertilizer usage significantly influence crop yield. Loamy soil with adequate rainfall and fertilizer gives the best crop production.